



Minimal FDs And BCNF Proofs

Minimal FD Sets & BCNF Proofs

Hotel / Transport / Holiday Package Booking System

Core Entities

- Vendor
- Hotel
- Hotel_Amenity
- Room_Category
- Room_Availability
- User
- Review
- Insurance

Transport

- Transport_Route
- Route_Schedule
- Schedule_Class
- Route_Schedule_Class
- Schedule_Departure

Packages

- Holiday_Package
- Package_Departure

Bookings

- Promo_Code
- Booking
- Hotel_Booking
- Guest_List
- Transport_Booking
- Passenger_List
- Package_Booking
- Package_Traveller
- Cancellation

Supporting

- Booking_Insurance
- Package_Hotel
- Package_Transport

Core Entities

Vendor

ATTRIBUTES: vendor_id, company_name, vendor_type, gst_number, registered_address, website, contact_name, contact_phone, contact_email, is_active, created_at

CANDIDATE KEY(S): {vendor_id}

MINIMAL FD SET (Canonical Cover):

vendor_id → company_name, vendor_type, gst_number, registered_address, website,
contact_name, contact_phone, contact_email, is_active, created_at

BCNF PROOF:

Candidate key: {vendor_id}.

The minimal FD set contains exactly one FD. Its determinant vendor_id is the sole candidate key, hence a superkey. Since every non-trivial FD has a superkey as its determinant, Vendor is in BCNF.

Hotel

ATTRIBUTES: hotel_id, hotel_name, city, full_address, star_rating, checkin_time, is_active, vendor_id

CANDIDATE KEY(S): {hotel_id}

MINIMAL FD SET (Canonical Cover):

hotel_id → hotel_name, city, full_address, star_rating, checkin_time, is_active, vendor_id

BCNF PROOF:

Candidate key: {hotel_id}.

The single non-trivial FD has hotel_id - the sole candidate key - as its determinant, making it a superkey. No FD in the minimal set violates BCNF. Hotel is in BCNF.

Note: the multi-valued attribute 'amenities' is factored out into Hotel_Amenity.

Hotel_Amenity

ATTRIBUTES: hotel id, amenity

CANDIDATE KEY(S): {(hotel_id, amenity)}

MINIMAL FD SET (Canonical Cover):

(No non-trivial FDs - the key spans the entire relation)

BCNF PROOF:

Candidate key: {hotel_id, amenity}.

The relation has only two attributes, both forming the key. There are no non-trivial FDs, so the condition for BCNF is satisfied vacuously. Hotel_Amenity is in BCNF.

Room_Category

ATTRIBUTES: room category id, hotel_id, category_name, max_guests, price_per_night, total_rooms

CANDIDATE KEY(S): {room_category_id}

MINIMAL FD SET (Canonical Cover):

room_category_id $\rightarrow$ hotel_id, category_name, max_guests, price_per_night, total_rooms

BCNF PROOF:

Candidate key: {room_category_id}.

The single FD's determinant is room_category_id, the sole candidate key and therefore a superkey. Room_Category is in *BCNF*.

Room_Availability

ATTRIBUTES: room category id, available date, available_rooms

CANDIDATE KEY(S): {(room_category_id, available_date)}

MINIMAL FD SET (Canonical Cover):

room_category_id, available_date → available_rooms

BCNF PROOF:

Candidate key: {room_category_id, available_date}.

The composite key is the only determinant that appears in the minimal FD set. A composite key is a superkey, so Room_Availability is in BCNF.

User

ATTRIBUTES: user_id, full_name, email, phone, dob, home_city, account_status

CANDIDATE KEY(S): {user_id} , {email}

MINIMAL FD SET (Canonical Cover):

user_id → full_name, email, phone, dob, home_city, account_status

email → user_id

BCNF PROOF:

Candidate keys: {user_id} and {email} (email is unique per user).

FD 1: determinant user_id is a candidate key (superkey).

FD 2: determinant email is a candidate key (superkey).

Both FDs in the minimal set have superkeys as determinants. User is in BCNF.

Review

ATTRIBUTES: review_id, user_id, entity_type, entity_id, star_rating, title, comment, reviewed_at

CANDIDATE KEY(S): {review_id}

MINIMAL FD SET (Canonical Cover):

$\text{review_id} \rightarrow \text{user_id}, \text{entity_type}, \text{entity_id}, \text{star_rating}, \text{title}, \text{comment}, \text{reviewed_at}$

BCNF PROOF:

Candidate key: {review_id}.

The determinant review_id is the sole candidate key, hence a superkey. Review is in BCNF.

Insurance

ATTRIBUTES: insurance_id, provider_name, plan_name, coverage_summary, premium_amount, valid_from, valid_to, claim_status

CANDIDATE KEY(S): {insurance_id}

MINIMAL FD SET (Canonical Cover):

$\text{insurance_id} \rightarrow \text{provider_name}, \text{plan_name}, \text{coverage_summary}, \text{premium_amount}, \text{valid_from}, \text{valid_to}, \text{claim_status}$

BCNF PROOF:

Candidate key: {insurance_id}.

The single FD's determinant is insurance_id, the sole candidate key and a superkey. Insurance is in BCNF.

Transport

Transport_Route

ATTRIBUTES: route_id, transport_mode, origin_city, destination_city, vehicle_number, vendor_id

CANDIDATE KEY(S): {route_id}

MINIMAL FD SET (Canonical Cover):

$\text{route_id} \rightarrow \text{transport_mode}, \text{origin_city}, \text{destination_city}, \text{vehicle_number}, \text{vendor_id}$

BCNF PROOF:

Candidate key: {route_id}.

route_id is the sole candidate key, so it is a superkey. Transport_Route is in BCNF.

Route_Schedule

ATTRIBUTES: schedule id, route_id, departure_time, arrival_time, days_of_week, is_active

CANDIDATE KEY(S): {schedule_id}

MINIMAL FD SET (Canonical Cover):

schedule_id $\rightarrow$ route_id, departure_time, arrival_time, days_of_week, is_active

BCNF PROOF:

Candidate key: {schedule_id}.

schedule_id is a superkey. Route_Schedule is in BCNF.

Schedule_Class

ATTRIBUTES: class id, class_name, total_seats, base_fare

CANDIDATE KEY(S): {class_id}

MINIMAL FD SET (Canonical Cover):

class_id $\rightarrow$ class_name, total_seats, base_fare

BCNF PROOF:

Candidate key: {class_id}.

class_id is the sole candidate key and a superkey. Schedule_Class is in BCNF.

Route_Schedule_Class

ATTRIBUTES: schedule id, class id

CANDIDATE KEY(S): {(schedule_id, class_id)}

MINIMAL FD SET (Canonical Cover):

(No non-trivial FDs - the key spans the entire relation)

BCNF PROOF:

Candidate key: {schedule_id, class_id}.

This is a pure junction relation. The key covers all attributes; no proper-subset FDs exist. BCNF holds vacuously.

Route_Schedule_Class is in BCNF.

Schedule_Departure

ATTRIBUTES: departure_id, schedule_id, class_id, departure_date, available_seats, dynamic_fare, status

CANDIDATE KEY(S): {departure_id}

MINIMAL FD SET (Canonical Cover):

departure_id $\rightarrow$ schedule_id, class_id, departure_date, available_seats, dynamic_fare, status

BCNF PROOF:

Candidate key: {departure_id}.

departure_id is the sole candidate key, hence a superkey. Schedule_Departure is in BCNF.

Packages

Holiday_Package

ATTRIBUTES: package_id, package_name, theme, duration_days, duration_nights, base_price_per_person, is_active, vendor_id

CANDIDATE KEY(S): {package_id}

MINIMAL FD SET (Canonical Cover):

package_id $\rightarrow$ package_name, theme, duration_days, duration_nights, base_price_per_person, is_active, vendor_id

BCNF PROOF:

Candidate key: {package_id}.

package_id is a superkey. Holiday_Package is in BCNF.

Package_Departure

ATTRIBUTES: package departure id, package_id, departure_dates, total_seats, available_seats, price_per_person, status

CANDIDATE KEY(S): {package_departure_id}

MINIMAL FD SET (Canonical Cover):

package_departure_id → package_id, departure_dates, total_seats, available_seats, price_per_person, status

BCNF PROOF:

Candidate key: {package_departure_id}.

package_departure_id is a superkey. Package_Departure is in BCNF.

Bookings

Promo_Code

ATTRIBUTES: promo id, code, discount_type, discount_value, max_discount_cap, min_booking_amt, valid_from, valid_to, applicable_on, used_count, is_active

CANDIDATE KEY(S): {promo_id}, {code}}

MINIMAL FD SET (Canonical Cover):

promo_id → code, discount_type, discount_value, max_discount_cap, min_booking_amt,
valid_from, valid_to, applicable_on, used_count, is_active
code → promo_id

BCNF PROOF:

Candidate keys: {promo_id} and {code} (promo codes are unique).

FD 1: determinant promo_id is a candidate key (superkey).

FD 2: determinant code is a candidate key (superkey).
Both determinants are superkeys. Promo_Code is in BCNF.

Booking

ATTRIBUTES: booking_id, user_id, booking_type, booked_at, status, base_amount, discount_amount, final_amount, payment_method, payment_ref, promo_id

CANDIDATE KEY(S): {booking_id}

MINIMAL FD SET (Canonical Cover):

booking_id → user_id, booking_type, booked_at, status, base_amount,
discount_amount,
final_amount, payment_method, payment_ref, promo_id

BCNF PROOF:

Candidate key: {booking_id}.
booking_id is the sole candidate key, hence a superkey. Booking is in BCNF.

Hotel_Booking

BCNF

ATTRIBUTES: booking_id, room_category_id, checkin_date, checkout_date,
num_rooms, num_guests

CANDIDATE KEY(S): {booking_id}

MINIMAL FD SET (Canonical Cover):

booking_id → room_category_id, checkin_date, checkout_date, num_rooms,
num_guests

BCNF PROOF:

Candidate key: {booking_id} (1-to-1 specialisation of Booking).
booking_id is the sole candidate key and a superkey. Hotel_Booking is in BCNF.

Guest_List

ATTRIBUTES: booking_id, guest_name, age, gender, govt_id_type, govt_id_number

CANDIDATE KEY(S): {(booking_id, guest_name)}

MINIMAL FD SET (Canonical Cover):

booking_id, guest_name $\rightarrow$ age, gender, govt_id_type, govt_id_number

BCNF PROOF:

Candidate key: {booking_id, guest_name} (weak entity identified within a booking).

The composite key is a superkey. The single FD's determinant equals the candidate key. Guest_List is in BCNF.

Transport_Booking

ATTRIBUTES: booking_id, departure_id, num_seats

CANDIDATE KEY(S): {booking_id}

MINIMAL FD SET (Canonical Cover):

booking_id $\rightarrow$ departure_id, num_seats

BCNF PROOF:

Candidate key: {booking_id} (1-to-1 specialisation of Booking).

booking_id is a superkey. Transport_Booking is in BCNF.

Passenger_List

ATTRIBUTES: booking_id, passenger_name, age, gender, govt_id_type, govt_id_number

CANDIDATE KEY(S): {(booking_id, passenger_name)}

MINIMAL FD SET (Canonical Cover):

booking_id, passenger_name $\rightarrow$ age, gender, govt_id_type, govt_id_number

BCNF PROOF:

Candidate key: {booking_id, passenger_name} (weak entity).
The composite candidate key is a superkey. Passenger_List is in BCNF.

Package_Booking

BCNF

ATTRIBUTES: booking_id, package_departure_id, num_travellers

CANDIDATE KEY(S): {booking_id}

MINIMAL FD SET (Canonical Cover):

booking_id → package_departure_id, num_travellers

BCNF PROOF:

Candidate key: {booking_id} (1-to-1 specialisation of Booking).

booking_id is a superkey. Package_Booking is in BCNF.

Package_Traveller

ATTRIBUTES: booking_id, traveller_name, age, gender, govt_id_type,
govt_id_number

CANDIDATE KEY(S): {(booking_id, traveller_name)}

MINIMAL FD SET (Canonical Cover):

booking_id, traveller_name → age, gender, govt_id_type, govt_id_number

BCNF PROOF:

Candidate key: {booking_id, traveller_name} (weak entity).

The composite candidate key is a superkey. Package_Traveller is in BCNF.

Cancellation

ATTRIBUTES: cancellation_id, booking_id, reason, cancelled_at, refund_amount,
refund_status, processed_at

CANDIDATE KEY(S): {cancellation_id}

MINIMAL FD SET (Canonical Cover):

cancellation_id → booking_id, reason, cancelled_at, refund_amount,
refund_status, processed_at

BCNF PROOF:

Candidate key: {cancellation_id}.

cancellation_id is a superkey. Cancellation is in BCNF.

Supporting

Booking_Insurance

ATTRIBUTES: booking id, insurance id

CANDIDATE KEY(S): {(booking_id, insurance_id)}

MINIMAL FD SET (Canonical Cover):

(No non-trivial FDs - the key spans the entire relation)

BCNF PROOF:

Candidate key: {booking_id, insurance_id}.

Pure junction relation. No proper-subset FDs exist. BCNF holds by default.

Package_Hotel

ATTRIBUTES: package id, hotel id

CANDIDATE KEY(S): {(package_id, hotel_id)}

MINIMAL FD SET (Canonical Cover):

(No non-trivial FDs - the key spans the entire relation)

BCNF PROOF:

Candidate key: {package_id, hotel_id}.

Pure junction relation. BCNF holds by default.

Package_Transport

ATTRIBUTES: package id, departure id

CANDIDATE KEY(S): {(package_id, departure_id)}

MINIMAL FD SET (Canonical Cover):

(No non-trivial FDs - the key spans the entire relation)

BCNF PROOF:

Candidate key: {package_id, departure_id}.

Pure junction relation. BCNF holds by default.